Homework 2: The Spatial Distribution of Farmers Markets in Philadelphia

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1) Introduction

In the modern day, it is widely accepted that having access to healthy and locally grown food is extremely beneficial in ensuring the health of people. Despite this, in recent times, it has been raised that access to this healthy, locally grown food has become a challenge for many residents of Philadelphia in the United States, particularly in neighborhoods with less access to farmers markets despite a campaign from the Philadelphia Food Trust to promote farmers markets. These markets provide many benefits, such as fresh, seasonal and healthy produce, give economic opportunity for local farmers and growers, and also provide a space for community interaction. However, these markets are not distributed equally across the city, with many neighborhoods seemingly completely devoid of any markets entirely such as the Northern and Southern areas of the city. This report seeks to investigate whether the spatial distribution of farmers markets across Philadelphia is random, clustered, or dispersed. Two spatial statistical tools will be used to examine this - the Nearest Neighbour Analysis and Ripley's K-Function Analysis. This analysis will be performed within ArcGIS Pro by evaluating the pattern of farmers markets in 2013, based on their geolocated positions and a zip-code map of Philadelphia city.

2) Methods

a) Hypotheses

i) Complete Spatial Randomness (CSR)

Complete Spatial Randomness assumes that every location in a study area has an equal chance of containing a point (in this case a farmers market), and that the location of one point does not influence the location of another. If CSR holds, the pattern of points will have no spatial structure, meaning that they will neither cluster with each other or be dispersed from each. In this study, CSR acts as our null model when we compare the distribution of farmers markets in Philadelphia.

ii) Null and Alternative Hypotheses

Null Hypothesis (H_0): The observed pattern is random. In this case, this would mean the farmers markets are distributed according to CSR.
Alternate Hypothesis (H_1): The observed pattern is NOT random. Farmers markets are not distributed randomly and instead show spatial clustering or dispersion.

b) Nearest Neighbour Analysis

i) Method Description

The Nearest Neighbour Analysis measures the distance from each point to its closest neighbouring point. It then calculates the average of those distances for all points, and then compares it to the average expected distance under CSR. This way we can determine whether farmer markets are more clustered together or more dispersed than our random expectation.

ii) Test Statistic

Our formula for the Nearest Neighbour Index is:

$$NNI = \frac{Observed\ Average\ Distance}{Expected\ Average\ Distance\ (when\ pattern\ is\ random)} = \frac{\bar{D}_O}{\bar{D}_E}$$

Here, the observed average distance is the average of the nearest-neighbour distances in our dataset. That is, the distance between farmer markets. Expected average distance assumes CSR as previously explained above.

When the NNI = 1, or close to it, it means that there is a random pattern in distribution. It implies that the observed average distance is equal to, or almost equal to the expected average distance under CSR.
When the NNI is closer to 0, that means we have a clustered pattern. Very close to 0, it would mean all of our points are located in the same area.
When the NNI is closer to 2 (maximum of 2.149), that means we instead have a dispersed pattern. This implies that the observed average distance is much higher than the expected average distance had the pattern been random.

We also calculate a Z-score and p-value to confirm the statistical significance of our results:

$$z = \frac{\bar{D}_O - \bar{D}_E}{SE}$$

Here, SE represents the standard error. If Z > 1.96 or Z < -1.96 then that would mean our results are statistically significant at 95% confidence. If Z > 1.96, we have significant dispersion, and if Z < -1.96 we have significant clustering. Both of these results mean that our p-value is < 0.05, which would allow us to reject the Null Hypothesis (H_0).]

iii) Limitations

There are, however, a number of limitations associated with the Nearest Neighbour Index. It only considers the nearest neighbours of points and ignores the broader patterns. Additionally, it is very sensitive to the shape and size of a study area. Whilst using a rectangular boundary box may include irrelevant space, an actual city boundary with a correctly calculated area space would give a more accurate picture.

c) Ripley's K-Function Analysis

i) Method Description

The K-function extends spatial analysis across multiple distance bands, in contrast to the NNI, making it suitable for identifying spatial patterns at different scales. For each distance (d), it calculates how many other points are within the radius of each point, compared to what would be expected under CSR.

$$K(d) = \frac{(\sum_{i=1}^{n} \#[S \in Cirlce(s_i, \boldsymbol{d})])/n}{n/a} = \frac{Mean\ \#\ points\ in\ all\ circles\ of\ radius\ \boldsymbol{d}}{Mean\ pt\ density\ in\ entire\ study\ region\ \boldsymbol{a}}$$

K(d) then represents the mean number of points you observe within a circle of radius d. It does this whilst also taking into account the overall point density in our study region, Philadelphia. However, generally, for the ease of interpretation, L(d) is instead used:

$$L(d) = \sqrt{\frac{K(d)}{\pi}} - d,$$

When L(d) = 0, it is within CSR.
When L(d) > 0, there is clustering at scale d
When L(d) < 0, there is dispersion at scale d

Note that ArcGIS Pro uses a slightly different form of this:

$$\boxed{Under\ CSR, \qquad L(d) = \sqrt{\frac{\pi d^2}{\pi}} = d}$$

L(d) > d: Clustering
L(d) < d: Dispersion

ii) Testing Procedure

In our analysis in ArcGIS Pro:

- 10 distance bands were used, starting from 0 and increasing by 2,500 feet each time up until 25,000 feet.
- 99 permutations of random patterns were generated to make confidence envelopes.
- If the observed K or L values fall outside of these envelopes then it means that the result is considered statistically significant.

iii) Near-Border Points

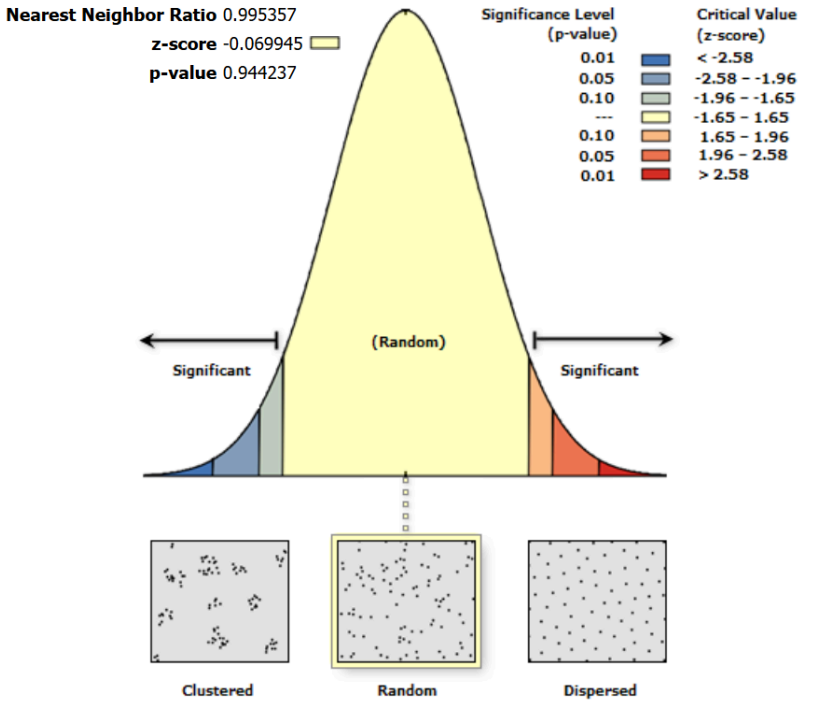
One other issue is that points near the edges of the city boundaries may appear to have fewer neighbours because no points lie beyond the city boundary in our data, which can skew the results. In order to account for this, we use the Simulate Outer Boundary Values Edge Correction function in ArcGIS Pro, which simulates possible neighbours outside of our boundary. Ripley's Edge Correction is another possible tool, but this generally works best with rectangles, squares or other simple shapes and not city boundaries.

3) Results

a) Nearest Neighbour Analysis

i) Using Minimum Bounding Rectangle

Average Nearest Neighbor Summary



Given the z-score of -0.069945, the pattern does not appear to be significantly different than random.

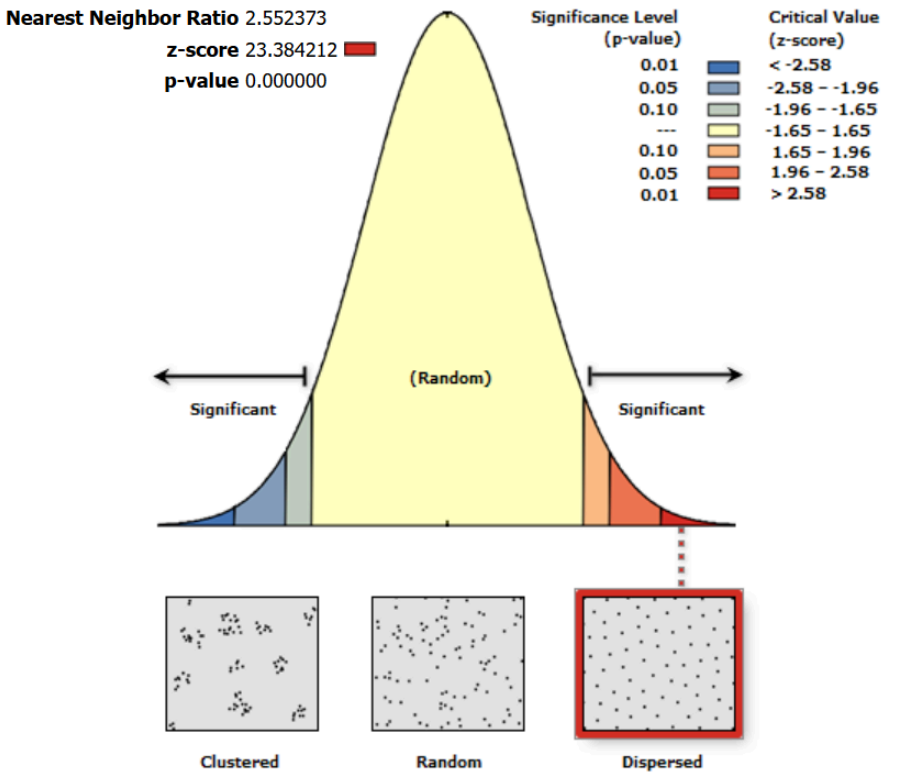
Average Nearest Neighbor Summary

Observed Mean Distance	3112.9097 feet
Expected Mean Distance	3127.4314 feet
Nearest Neighbor Ratio	0.995357
z-score	-0.069945
p-value	0.944237

Here, we see that the observed mean distance is only slightly less than what would be expected under randomness. With a nearest neighbour ratio of 0.995, being very close to 1, it would indicate that the spacing between the markets is practically identical to random distribution. A p-value of 0.944 is above our 0.05 threshold for statistical significance, so we fail to reject the null hypothesis. Based on these results, we would conclude that the farmers market distribution seems random. However, as mentioned previously, since this uses a minimum bounding triangle this is likely misleading.

ii) Using Actual Area of City

Average Nearest Neighbor Summary



Given the z-score of 23.384212, there is a less than 1% likelihood that this dispersed pattern could be the result of random chance.

Average Nearest Neighbor Summary

Observed Mean Distance	3112.9097 feet
Expected Mean Distance	1219.6140 feet
Nearest Neighbor Ratio	2.552373
z-score	23.384212
p-value	0.000000

Here, we get a totally different result. The expected mean distance is now much lower when calculated using Philadelphia's actual size, whilst the average observed distance remains the same. It is now more than twice as far as would be expected under CSR. The Nearest Neighbour value is now at 2.55, suggesting significant spatial dispersion, whilst our p-value is now at 0 - meaning we strongly reject the null hypothesis. According to this examination, farmers markets are not randomly placed but are in fact dispersed implying they tend to be intentionally spaced out. However - we still have one more form of analysis to perform.

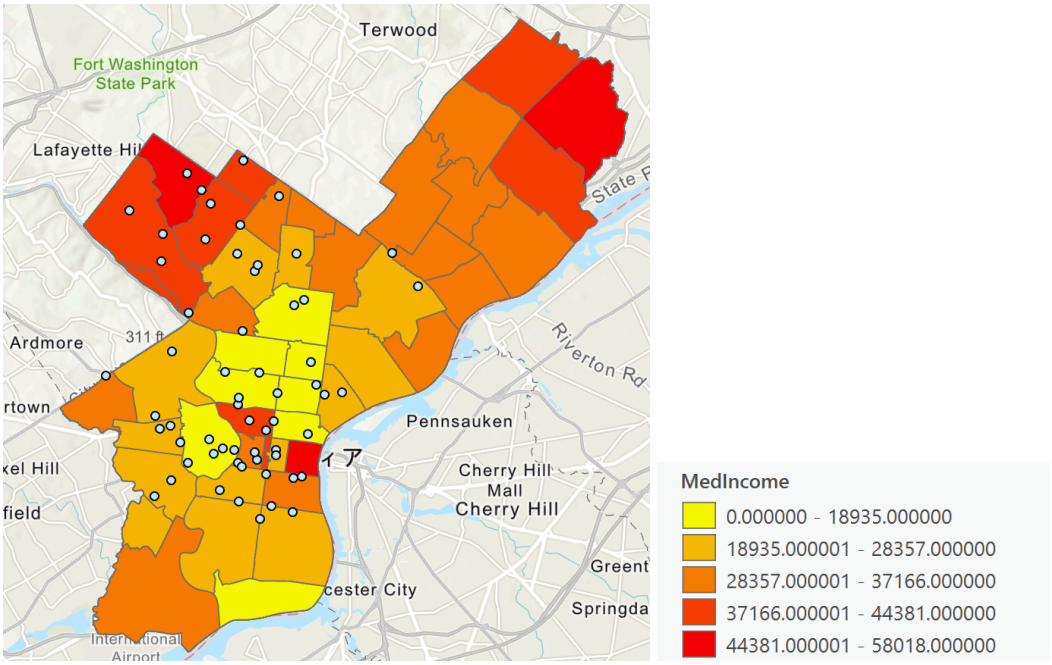
b) K-Function Analysis

i) Results

	OBJECTID *	ExpectedK	ObservedK	DiffK	LwConfEnv	HiConfEnv
1	1	2500	3701.59289	1201.59289	2163.02103	4248.088971
2	2	5000	7691.009666	2691.009666	4551.898493	6888.757489
3	3	7500	11959.653809	4459.653809	7032.788901	9213.265411
4	4	10000	15863.32464	5863.32464	9827.558673	11959.653809
5	5	12500	19372.541598	6872.541598	12317.558978	14578.917616
6	6	15000	22737.45648	7737.45648	14873.929501	17265.499656
7	7	17500	25930.489319	8430.489319	17400.46374	19815.984655
8	8	20000	28846.717239	8846.717239	19510.059147	22194.505115
9	9	22500	31334.48922	8834.48922	21560.572522	24662.234219
10	10	25000	33454.481074	8454.481074	23602.854212	27170.309273
	Click to add new row.					

Here, we can see that at every distance starting from 5,000ft, the ObservedK or L(d), was above the upper confidence envelope (HiConfEnv). This indicates statistically significant clustering at those distances. It was consistently higher than what would occur randomly, implying that farmers markets are spatially clustered at these distances The fact that this is not shown at just one scale but at multiple implies that there is certainly a clustering pattern in farmers markets, and that this is not a local anomaly, leading us to also strongly reject the Null Hypothesis.

This directly contradicts the dispersion conclusion of our previous Nearest Neighbour Analysis, which highlights our previously mentioned limitations of the NNA, and the importance of having multi-scales within the K-function. This also lines up far more accurately with what we can see visually on our map, which shows that the central and west neighbourhoods have many markets whilst in the north there are large empty areas.



ii) Population Explanation

Areas in the Northern and Southern parts of Philadelphia do indeed show few or no farmers markets. However, on visual inspection, this does not seem to be explained by income differences. After all, the northern areas show high median incomes in comparison to some of the central areas of Philadelphia with comparatively lower median incomes. Thus, a far more logical explanation would be a lower population and lower population density in these areas.

As a farmers market requires enough customers to be sustainable, it is quite plausible that the markets are so centrally located as it is the only location they see enough foot traffic to remain profitable. Note also that it is likely that the vast majority of those living in Northern and Southern areas commute directly into the city centre for work etc, which could reduce the need for any markets in North and South Philadelphia. Regardless however, the pattern would still remain just as clustered as it is now - just it would be more justified by the population difference. It would not change the fact that they are heavily clustered into Central and Western Philadelphia.

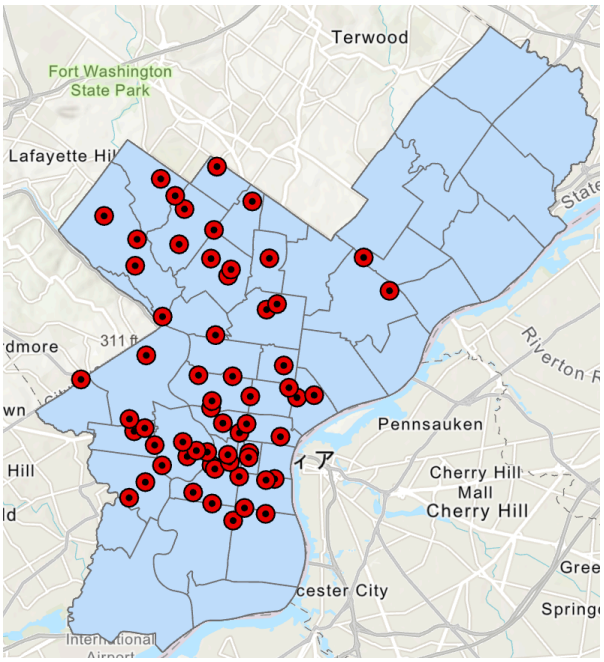
4) Discussion

a) Are NNA and K-Function Results Consistent?

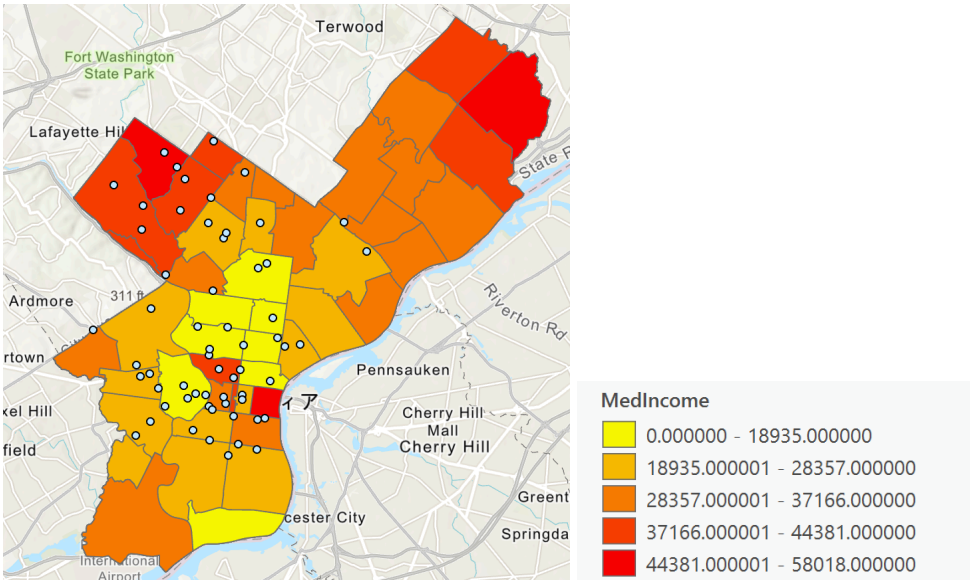
No - they directly contradict each other with our NNA (using our city boundary) showing us dispersion whilst our K-function shows clustering. However, as previously mentioned, this is because the NNA is more limited and is only able to detect patterns based on its nearest neighbour.

b) Are the results consistent with visual expectations?

Yes, they are. Our map clearly shows that there are large clusters in the Central and Western parts of Philadelphia whilst the Northern and Southern areas are practically barren. This matches up more with our K-function analysis.



c) Effects of Income



Without doing any statistical tests, it is difficult to measure the influence of income on the prevalence of farmers markets. However, visually, this does not appear to have a massive effect. Whilst it is true that the Western neighbourhoods have a larger amount of markets whilst being wealthy, the similarly wealthy Northern neighborhoods do not have any. Additionally, the Central neighbourhoods also have a large number of markets despite their lower median income.

d) Final Conclusion

Relying on the K-function, we can conclude that the farmers markets are significantly clustered within Philadelphia. The pattern is not random, it shows clear spatial concentration. The reasons for this can be debated, but since income does not seem to be the main factor, we can assume it has to do with population density and the feasibility of operating a market in less dense areas.

e) Policy Implications

Whatever the justification, it is clear that the clustering of farmer markets has resulted in some neighbourhoods and areas of the city being completely excluded from access to healthy and fresh produce, which is a key issue for the Philadelphia Food Trust to tackle. Incentives should be given to help support markets in underserved areas, whether that be because of a lack of foot traffic due to low population density or due to a lack of income in the area.